

Exercise Set 4

Quantitative Trading and Price Impact
MSc in Mathematics and Finance

2025-2026

- (a) **Implementing a backtest function.** Given an intraday midprice series $(P_t)_t$ and the corresponding signed traded volume series $(q_t)_t$, implement a function that returns the impact-distorted price series under a fixed impact model (linear or square-root).

As inputs, use the impact parameters (λ, β) together with the daily volatility estimate σ and daily traded volume estimate ADV for the stock. The function should

- (i) transform and scale the signed volumes according to the chosen impact model and scales, and apply an exponential moving average with mean-reversion parameter β to obtain an intraday impact state as in Exercise 2(b),
 - (ii) add λ times this impact state to the cumulative return of the original price series and convert back to prices.
- (b) **Applying the backtest function.** Using the function from (a), consider AAPL on 21 Feb 2019 and run the following analysis for both impact model types (linear and square-root), using the corresponding fitted λ for AAPL from Exercise 3(a) and a half-life of 1 hour.

1. Using the signed traded volume series from the public tape as input trades, compute the impact-free prices by applying the backtest function with the *negative* of the public order flow (i.e., removing the estimated impact of the tape from the observed prices). Plot observed prices and impact-free prices for both model types.
2. On the impact-free price path, simulate the following buy programs:
 - (i) TWAP buy: trade a constant amount in each 10-second bin so that the total volume equals 20% of ADV.
 - (ii) VWAP buy: trade proportionally to the absolute public traded volume in each bin, i.e., trade 0.2 times the public volume profile.

For each strategy and model type, compute and plot:

1. the impact-free and the impact-distorted price path,
2. the cumulative impact (difference between resultant and impact-free prices),
3. the trade size per bin as a percentage of ADV, and
4. the cumulative naive P&L and realistic P&L (both as a percentage of cost basis), where naive P&L ignores price impact and realistic P&L uses the impact-distorted execution prices.